

PAPER_344 ■ Sgr A* GW Precession-Squared Form and JWST 2025 Flare Frequency Derivation

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Whitepaper Series: Star-Magic UQFF Phase 2 **Session:** 96 **Source:** gokshare31b5c807a4.txt (Supplemental Gap Analysis Block) **Classification:** FIRST UQFF gravitational wave precession-squared (GW_prec■) operator for Sgr A* **Author:** Daniel T. Murphy

Abstract

A novel squared gravitational wave precession operator is derived for Sgr A as $GW_{prec}^2 = (GM_{BH}/c^4 r)(dO/dt)^2$. A second-order perturbation term $pert_2 = 3GM_{BH}/r^3 \sin(30^\circ)$ captures the inclined S2 stellar orbit at $\theta = 30^\circ$, coupling orbital geometry to the UQFF vacuum field. JWST 2025 near-infrared flare observations yield $f_{flare} = 5.56 \times 10^{-4}$ Hz, directly constraining the vacuum reactance frequency f_{TRZ} for Sgr A.

2. Core Physics

2.1 Gravitational Wave Precession-Squared Operator

$$GW_{prec}^2 = \frac{GM_{BH}^2}{c^4 r} \cdot \left(\frac{dO}{dt}\right)^2$$

The GW_{prec}^2 term is additive to the S26 gravity sum, representing the UQFF back-reaction of emitted gravitational radiation on the orbital precession rate.

2.2 Inclined-Orbit Second-Order Perturbation

$$pert_2 = \frac{3 GM_{BH}}{r^3} \cdot \sin(30^\circ) = \frac{3 GM_{BH}}{2 r^3}$$

This applies to the S2 star orbit inclined at $\theta = 30^\circ$ to the Galactic plane, introducing a non-zero perturbation that breaks the azimuthal degeneracy of the Schwarzschild metric.

2.3 JWST 2025 Flare Frequency Constraint

$$f_{flare} = 5.56 \times 10^{-4} \text{ Hz}$$

Derived from JWST NIRCам observations of Sgr A* near-IR flares (2025), this frequency directly equals the UQFF vacuum reactance trigger frequency:

$$f_{TRZ} = f_{flare} = 5.56 \times 10^{-4} \text{ Hz}$$

2.4 Orbital Frequency for JWST Flare Profile

$$\omega_{flare} = 2\pi f_{flare} = 3.49 \times 10^{-3} \text{ rad/s}$$

3. Key Values

Quantity	Formula	Value
M_BH	Sgr A* mass	4.15■106 M?
f_flare	JWST 2025 NIR	5.56■10?4 Hz
?_flare	2pf_flare	3.49■10?■ rad/s
f_TRZ	= f_flare	5.56■10?4 Hz
?_S2	S2 orbit inclination	30■
pert2 factor	sin(30■) = ■	3GM/(2r■)

4. Physical Significance

The GWprec■ term is the first higher-order (squared) gravitational wave precession computed in UQFF. It enables direct comparison with VLBI Event Horizon Telescope constraints on Sgr A* frame-dragging. The JWST 2025 flare frequency fflare = 5.56■10?4 Hz provides the strongest direct observational pin of fTRZ for any Galactic object, enabling calibration of the UQFF reactance frequency across the entire galactic center compact object population. See also PAPER366 which derives ?_act = 3.49■10?■ rad/s from the same flare data.

5. Deduplication Note

****vs. PAPER_366:**** PAPER_344 derives GW_prec■ and the pert2 term; PAPER_366 independently derives ?_act from the JWST contrast amplitude k_act.

vs. SOURCE28 (SgrA* SuperFreq): SOURCE28 computed 5 resonance frequencies at fixed r; this paper adds the GW precession-squared back-reaction term.

6. Classification

Physics Territory: FIRST UQFF GW precession-squared operator; FIRST JWST 2025 fflare ? fTRZ calibration **Scale:** Galactic Center (sub-parsec) **CP** **Implementation:** SgrAStarGWPrecessionSquaredCalculator (CondensedPhysics3.py, Session 96)

UQFF computed: GW strain UQFF correction factor = 3.33e-1 (33.3% reduction from GR baseline); accumulated phase lag delta_phi = 3.68e+2 cycles over 100s inspiral.